CALVIN GACHERU MWANGI

Data Scientist | AI Engineer | Data Analyst

Location: Nairobi, Kenya **Email:** calvin.g.mwangi@gmail.com **LinkedIn:** linkedin.com/in/calvin-gacheru-057672281 **GitHub:** github.com/Calvin-Gacheru

PROFESSIONAL SUMMARY

Results-driven Data Science and Analytics student specializing in the complete machine learning lifecycle, from data engineering and model training to robust cloud deployment. Proven ability to build and ship production-ready Agentic AI systems, RAG architectures, and predictive machine learning models. Adept at bridging the gap between raw data and scalable, functional software, delivering end-to-end commercial solutions with a strong focus on automation, MLOps, and actionable business intelligence.

EDUCATION

Bachelor of Science in Data Science & Analytics *United States International University — Africa (USIU-Africa)* | Nairobi, Kenya *Expected Graduation: 2027 (Currently Year 4 of 4)*

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), SQL, R, Bash, HTML/CSS/JavaScript
- **Machine Learning & AI Stack:** scikit-learn, Pandas, NumPy, statsmodels, NLP, Time Series Analysis, Predictive Analytics
- **Generative AI & LLMs:** LangChain, LangGraph, RAG Systems, OpenAI API, Groq LLM, Agentic Orchestration, Vector Databases (Qdrant)
- **Data Engineering & Analytics:** ETL Pipelines, Exploratory Data Analysis, Data Visualization, Power BI, Matplotlib
- **Infrastructure & MLOps:** Docker, Kubernetes, CI/CD (GitHub Actions), Linux/Fedora, Git/GitHub, PostgreSQL, Redis
- **Web Frameworks:** FastAPI, Flask, Django
- **Languages:** English (Native), Swahili (Native)

PROFESSIONAL EXPERIENCE

Data Science Virtual Intern | *ProdigyInfoTech* | 2025

- Executed 4 end-to-end data science projects encompassing exploratory data analysis, predictive modeling, customer churn analytics, and NLP sentiment analysis.
- Developed reproducible data workflows and documented analytical insights derived from real-world datasets, emphasizing data quality and model interpretability.

Freelance Full-Stack Developer | *BikBat Insurance Agency* | 2026

- Engineered and launched a comprehensive corporate web platform end-to-end for a commercial insurance client.
- Managed the complete deployment lifecycle, including domain configuration, DNS management, and secure cloud hosting.

SELECTED DATA SCIENCE & AI PROJECTS

MySaidi - AI Productivity Assistant (Agentic AI & Web App)

- Architected a full-stack Progressive Web App (PWA) that leverages natural language processing to manage user schedules and habits.
- Designed a decoupled browser-to-REST API utilizing FastAPI and a PostgreSQL database.
- Integrated a LangGraph-powered AI agent driven by Groq LLMs to parse intents and dynamically query/modify user data via bound tool calling.

Uber Price Prediction & Production MLOps Pipeline

- Engineered an enterprise-grade MLOps architecture to predict ride prices using an optimized Random Forest Regressor.
- Automated the machine learning lifecycle through a GitHub Actions CI/CD pipeline and multi-container Docker orchestration.
- Deployed the model via a robust FastAPI REST backend with Pydantic validation, serving predictions to an interactive Streamlit UI.

Diabetes Risk Screening Tool (Predictive Analytics)

- Developed a recall-optimized diagnostic screening pipeline utilizing an XGBoost classifier, addressing severe class imbalance via SMOTE and algorithmic class weighting.
- Achieved an 88% recall rate for minority classes, prioritizing the minimization of false negatives in medical diagnostics.
- Deployed the pipeline as a Streamlit application featuring real-time automated feature engineering, binning, and data scaling.

Maternal Risk Detection System (Explainable AI)

- Built a predictive system for maternal health risks (Preeclampsia, Gestational Diabetes, Anemia) utilizing real-time patient vitals.
- Integrated SHAP (SHapley Additive exPlanations) to provide transparent model interpretability for clinical decision-making.
- Implemented automated task scheduling (APScheduler) and third-party API integration (Africa's Talking) to deliver SMS appointment reminders and wellness tips.

Behavioural Explainer (RAG Architecture)

- Constructed a production-grade Retrieval-Augmented Generation (RAG) system to analyze complex psychological framework scenarios.

- Engineered a document ingestion pipeline using PyMuPDF and text chunking, storing embeddings in a Qdrant vector database.
- Integrated local LLMs (Ollama, Llama 3.1) and developed a Command Line Interface (CLI) tool using Typer within a containerized Docker environment.

Lung Cancer Survival Prediction (Survival Analysis & Data Mining)

- Collaborated on a machine learning pipeline extracting, transforming, and analyzing clinical/treatment records to estimate survival outcomes.
- Engineered advanced clinical features, including temporal tracking and comorbidity scoring, to feed a Random Forest model.
- Developed an interactive Plotly dashboard to serve real-time survival probability estimations and clinical visual insights.

Stroke Prediction Using Bayesian Networks (Probabilistic Modeling)

- Developed a discrete Directed Acyclic Graph (DAG) Bayesian Network using `pgmpy` to estimate stroke risk via probabilistic reasoning.
- Performed exact inference via Variable Elimination and structural evaluation using BIC and BDeu scoring.

Twitter Entity Sentiment Analysis (NLP)

- Built a natural language processing pipeline to detect public sentiment towards specific brands on X (formerly Twitter).
- Utilized GridSearchCV for extensive hyperparameter tuning, achieving a 93% F1-score with the optimal classifier, and deployed the solution via Streamlit.

CERTIFICATIONS

- **Data Science Virtual Internship** – *ProdigyInfoTech (2025)*
- **Building AI Agents using LangGraph** – *Simplilearn*